

localparam

A parameter is a compile-time constant the parent can override at instantiation with hash-dot syntax

Public knob, private derivation

- Declare parameter WIDTH equals eight as the tunable bus width
- Then localparam DEPTH equals WIDTH times two
- Instantiate fifo hash-dot WIDTH open-paren four close-paren and WIDTH becomes four while
- Hash-dot DEPTH open-paren ninety-nine close-paren is illegal or ignored, DEPTH is local
- The pattern parameter W

Browser lab

Digital Design and Verification Platform

HomeTools

Home / Tools / localparam vs parameter

INTERACTIVE TOOL

localparam vs parameter

parameter is overridable at the instance. localparam is local — and why defparam is a bad habit.

Starter example: parameter WIDTH = 8 and localparam DEPTH = WIDTH*2. Override WIDTH to 4 via #(.WIDTH(4)) — DEPTH becomes 8. You cannot override DEPTH.

Load starter example

Challenges (1/22)

Quiz: parameter. Overridable module constant is a? Answer: parameter

Your answer

Show hintCheckNextIdle

quiz-paramquiz-localquiz-defparamquiz-preferstarteroverride-4apply-bncleartry-local

defparam-showexplainderivedquiz-deriveddefault-changecode-has-localparamcode-poundquiz-track

warn-localquiz-publicno-defparam-preferdepth-formulafull-path

Core ideas

parameter

Public constant — instance may override with #().

localparam

Private / derived — not overridable from outside.

defparam

Hierarchical poke — deprecated; prefer #().

Module & instance

Elaborated values

localparam lab starter

Real Verilog practice

```
wsl - module11-localparam-lab/examples/navigation

# Track A - real Unix

# real Verilog session (Track A)

$ cat examples/localparam_demo/localparam_demo.v
// localparam_demo.v - parameter vs localparam
module fifo #(
    parameter WIDTH = 8
)(
    input  [WIDTH-1:0] d,
    output [WIDTH-1:0] q
);
    localparam DEPTH = WIDTH * 2; // derived - not overridable
    assign q = d;
endmodule

module top;
    wire [3:0] din, dout;
    fifo #(.WIDTH(4)) u (.d(din), .q(dout));
endmodule

$ iverilog -t null examples/localparam_demo/localparam_demo.v
(syntax check passed)
```

Real Verilog — parameter and localparam in one module

Pitfalls to watch

- Do not put every constant in parameter form
- Do not use defparam for new code; prefer hash-dot at the instance
- Do not assume a localparam override fails loudly in every tool
- Do keep parameter for instance-visible knobs and localparam for math that must stay in

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, override WIDTH to four and state the new DEPTH
- On real Verilog, add a localparam derived from an existing parameter
- When you are ready